Section 3.1 IM-OLS Cointegration Estimation Matrix: U.S. NFC Transformation Elasticity (1929–2024)

Department of Economics, UMass Amherst

August 2026

Data Provenance and Estimation Summary

This report presents the empirical cointegration results for Section 3.1 of Chapter 2. All models are estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) over the full NIPA sample window (1929–2024, $N = 95$ effective differenced observations). Point estimates are tuning-parameter free; fixed-b HAR t-statistics use a Bartlett kernel with Newey-West bandwidth. Distributional conditioning variables (ω_{t-1} and ι_{t-1}) are centered at their 1929 initial values, preserving $\hat{\theta}_0$ as the capital elasticity at the initial state. VIF statistics are computed on the level regressors space. To ensure robustness against structural instability, each specification dynamically incorporates step level-shift dummies $D_{T_b}^{\text{step}}$ for structural breaks identified by a first-pass Bai-Perron residual audit.

1 Specification Set 1: Double Interactions ($k^{\text{cap}} \cdot \omega_{t-1}$)

Table 1: Specification Set 1: Double Interaction IM-OLS Cointegration Estimates (1929–2024)

Dependent Variable: y_t^{NFC} (Log NFC Real GVA)						
Model	M1.1	M1.2	M1.3	M1.4	M1.5	M1.6
Capital Stock	k_1^{cap}	k_2^{cap}	k_1^{cap}	k_2^{cap}	k_1^{cap}	k_2^{cap}
Demand Control	None	None	None	None	None	None
$\hat{\alpha}$ (Intercept)	8.9612***	6.8378***	9.1933***	8.7403***	5.4203***	4.7989***
	(0.8316)	(0.7823)	(0.9077)	(0.8247)	(1.1418)	(1.1492)
$\hat{\gamma}_{1944}^{\text{step}}$ (D_{1944}^{step})	0.5742***	0.3895***	—	0.5825***	—	—
	(0.1282)	(0.1157)		(0.1211)		
$\hat{\gamma}_{1945}^{\text{step}}$ (D_{1945}^{step})	—	—	0.5611***	—	—	—
			(0.1357)			
$\hat{\gamma}_{1965}^{\text{step}}$ (D_{1965}^{step})	0.3886***	0.1043	0.4002***	0.3607***	—	—
	(0.1329)	(0.1282)	(0.1420)	(0.1301)		
$\hat{\gamma}_{1967}^{\text{step}}$ (D_{1967}^{step})	—	—	—	—	—	-
						0.3774***
						(0.1461)
$\hat{\gamma}_{1968}^{\text{step}}$ (D_{1968}^{step})	—	—	—	—	-0.3383**	—
					(0.1484)	
$\hat{\gamma}_{1981}^{\text{step}}$ (D_{1981}^{step})	—	—	—	—	—	-0.3029**
						(0.1341)
$\hat{\gamma}_{1982}^{\text{step}}$ (D_{1982}^{step})	—	0.0053	—	—	-0.2665**	—
		(0.1307)			(0.1342)	
$\hat{\gamma}_{1985}^{\text{step}}$ (D_{1985}^{step})	0.2514**	—	0.2818**	0.2511**	—	—
	(0.1243)		(0.1338)	(0.1177)		
$\hat{\gamma}_{1996}^{\text{step}}$ (D_{1996}^{step})	—	—	—	—	0.1164	0.0712
					(0.1030)	(0.1010)
$\hat{\gamma}_{2003}^{\text{step}}$ (D_{2003}^{step})	0.1915	—	0.1837	0.1591	—	—
	(0.1622)		(0.1639)	(0.1491)		
$\hat{\gamma}_{2010}^{\text{step}}$ (D_{2010}^{step})	—	—	—	—	-	-
					0.3135***	0.3749***
					(0.1032)	(0.1025)
$\hat{\theta}_0$ (Capital Elasticity)	0.3447***	0.5192***	0.3276***	0.3545***	0.6684***	0.7046***
	(0.0731)	(0.0670)	(0.0799)	(0.0710)	(0.0978)	(0.0961)
$\hat{\theta}_1$ (Interaction)	-0.0526	-0.0711	-0.0733	-0.0394	-0.1963	-0.1834
	(0.1415)	(0.0960)	(0.1464)	(0.1289)	(0.1225)	(0.1174)
Collinearity Diagnostics (VIF)						
VIF (k^{cap})	1.38	1.40	1.54	1.56	8.04	8.14
VIF (Interaction)	1.38	1.40	1.54	1.56	8.04	8.14
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.261	-1.725	-2.556	-2.128	-4.685***	-4.506***
Shin LM (C_{Shin}) [H_0: Coint.]	0.631***	1.024***	0.521***	0.707***	0.236***	0.225***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	5.532	802.808***	5.921	2750.381***	2585.040***	1472.400***
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
Sample Size (N)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$ based on fixed-b HAR t-statistics (Bartlett kernel, Newey-West bandwidth). **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0). Step dummies $D_{T_b}^{\text{step}} = \mathbb{I}(t \geq T_b)$ represent persistent level shifts from structural breaks identified via a first-pass Bai-Perron residual audit (15% trimming), distinguished from impulse outlier controls $D_{T_b}^{\text{imp}} = \mathbb{I}(t = T_b)$.

2 Specification Set 2: Triple Interactions ($k^{\mathbf{cap}} \cdot \omega_{t-1} \cdot \iota_{t-1}$)

Table 2: Specification Set 2 (Part A: Baseline NFC & Corporate GVA Wage Shares): Triple Interaction IM-OLS Cointegration Estimates

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.1	M2.2	M2.3	M2.4	M2.5	M2.6
Capital Stock	k_1^{cap}	k_1^{cap}	k_2^{cap}	k_2^{cap}	k_1^{cap}	k_1^{cap}
Demand Control	None	NetC Share	None	NetC Share	None	NetC Share
$\hat{\alpha}$ (Intercept)	6.1939***	7.3855***	6.3826***	7.8864***	6.3121***	7.3323***
	(1.0389)	(1.0486)	(0.9336)	(0.9797)	(1.0841)	(1.0804)
$\hat{\gamma}_{1944}^{\text{step}}$ (D_{1944}^{step})	0.2744**	0.3794**	0.3307***	0.5467***	0.2883**	0.3747**
	(0.1373)	(0.1680)	(0.1256)	(0.1605)	(0.1388)	(0.1683)
$\hat{\gamma}_{1965}^{\text{step}}$ (D_{1965}^{step})	—	—	0.0404	0.2039*	—	—
			(0.1439)	(0.1157)		
$\hat{\gamma}_{1966}^{\text{step}}$ (D_{1966}^{step})	0.0031	0.0893	—	—	0.0188	0.0866
	(0.1612)	(0.1257)			(0.1631)	(0.1275)
$\hat{\gamma}_{1981}^{\text{step}}$ (D_{1981}^{step})	—	-0.0694	—	0.0614	—	-0.0728
		(0.1392)		(0.1240)		(0.1419)
$\hat{\gamma}_{1982}^{\text{step}}$ (D_{1982}^{step})	-0.1213	—	-0.0579	—	-0.1088	—
	(0.1644)		(0.1467)		(0.1673)	
$\hat{\gamma}_{1998}^{\text{step}}$ (D_{1998}^{step})	—	—	—	0.2741***	—	—
				(0.0955)		
$\hat{\gamma}_{1999}^{\text{step}}$ (D_{1999}^{step})	—	0.2478**	—	—	—	0.2427**
		(0.1042)				(0.1042)
$\hat{\theta}_0$ (Capital Elasticity)	0.5910***	0.5186***	0.5599***	0.4377***	0.5804***	0.5230***
	(0.0904)	(0.0807)	(0.0795)	(0.0735)	(0.0943)	(0.0832)
$\hat{\theta}_1$ (Interaction)	-0.0938	0.8792	0.0073	0.5165	-0.0809	0.7677
	(1.0023)	(0.8449)	(0.9233)	(0.7511)	(0.9682)	(0.8255)
$\hat{\beta}_{\text{cm}}$ ($\text{NetC}^{\text{share}}$)	—	-0.7320	—	-0.3350	—	-0.7244
		(0.7361)		(0.7181)		(0.7359)
Collinearity Diagnostics (VIF)						
VIF (k^{cap})	1.57	2.54	1.59	2.61	1.74	2.79
VIF (Interaction)	1.57	1.63	1.59	1.67	1.74	1.82
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.051	-1.666	-1.956	-1.876	-2.076	-1.629
Shin LM (C_{Shin}) [H_0: Coint.]	0.781***	0.770***	0.867***	0.667***	0.770***	0.771***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	1542.222***	19074.593***	1947.135***	21688.445***	2222.232***	19123.867***
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
Sample Size (N)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$ based on fixed-b HAR t-statistics (Bartlett kernel, Newey-West bandwidth). **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0). Step dummies $D_{T_b}^{\text{step}} = \mathbb{I}(t \geq T_b)$ represent persistent level shifts from structural breaks identified via a first-pass Bai-Perron residual audit (15% trimming), distinguished from impulse outlier controls $D_{T_b}^{\text{imp}} = \mathbb{I}(t = T_b)$.

Table 3: Specification Set 2 (Part B: Shaikh-Tonak Tier 2 Wage Share): Triple Interaction IM-OLS Cointegration Estimates

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.7	M2.8	M2.9	M2.10	M2.11	M2.12
Capital Stock	k_2^{cap}	k_2^{cap}	k_1^{cap}	k_1^{cap}	k_2^{cap}	k_2^{cap}
Demand Control	None	NetC Share	None	NetC Share	None	NetC Share
$\hat{\alpha}$ (Intercept)	6.5185***	7.0269***	9.4698***	10.3466***	9.3943***	10.0377***
	(0.9753)	(1.1053)	(1.2651)	(0.7586)	(1.2349)	(0.8491)
$\hat{\gamma}_{1944}^{\text{step}}$ (D_{1944}^{step})	0.3451***	0.4292***	0.5366***	0.3764***	0.5388***	0.4024***
	(0.1270)	(0.1612)	(0.1424)	(0.1129)	(0.1336)	(0.1148)
$\hat{\gamma}_{1965}^{\text{step}}$ (D_{1965}^{step})	0.0583	—	—	—	—	—
	(0.1456)					
$\hat{\gamma}_{1966}^{\text{step}}$ (D_{1966}^{step})	—	0.0751	—	—	0.3443**	—
		(0.1275)			(0.1486)	
$\hat{\gamma}_{1967}^{\text{step}}$ (D_{1967}^{step})	—	—	0.3447**	0.0225	—	0.0083
			(0.1617)	(0.0825)		(0.0871)
$\hat{\gamma}_{1981}^{\text{step}}$ (D_{1981}^{step})	—	-0.0545	—	0.0372	—	0.0536
		(0.1339)		(0.0933)		(0.0953)
$\hat{\gamma}_{1982}^{\text{step}}$ (D_{1982}^{step})	-0.0430	—	0.1010	—	0.1426	—
	(0.1491)		(0.1517)		(0.1448)	
$\hat{\gamma}_{1996}^{\text{step}}$ (D_{1996}^{step})	—	—	—	-0.1577**	—	-0.1629**
				(0.0798)		(0.0821)
$\hat{\gamma}_{1998}^{\text{step}}$ (D_{1998}^{step})	—	0.2276**	—	—	—	—
		(0.0981)				
$\hat{\gamma}_{2010}^{\text{step}}$ (D_{2010}^{step})	—	—	—	-0.4853***	—	-0.4974***
				(0.0923)		(0.0950)
$\hat{\theta}_0$ (Capital Elasticity)	0.5480***	0.5232***	0.3020***	0.3148***	0.3013***	0.3239***
	(0.0830)	(0.0827)	(0.1112)	(0.0627)	(0.1060)	(0.0676)
$\hat{\theta}_1$ (Interaction)	0.0032	0.8710	-2.2463**	-5.5806***	-2.0137**	-5.3411***
	(0.8962)	(0.7816)	(0.9038)	(0.9425)	(0.8813)	(0.9721)
$\hat{\beta}_{\text{cm}}$ ($\text{NetC}^{\text{share}}$)	—	-0.5176	—	-1.6602***	—	-1.4971***
		(0.7106)		(0.4394)		(0.4513)
Collinearity Diagnostics (VIF)						
VIF (k^{cap})	1.78	2.88	6.76	9.56	7.09	10.32
VIF (Interaction)	1.78	1.87	6.76	7.53	7.09	8.07
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.003	-1.708	-2.156	-2.972	-2.213	-2.954
Shin LM (C_{Shin}) [H_0: Coint.]	0.847***	0.704***	0.515***	0.124**	0.501***	0.132**
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	2740.333***	19999.662***	202.861***	36650.597***	161.618***	36740.006***
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
Sample Size (N)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$ based on fixed-b HAR t-statistics (Bartlett kernel, Newey-West bandwidth). **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0). Step dummies $D_{T_b}^{\text{step}} = \mathbb{I}(t \geq T_b)$ represent persistent level shifts from structural breaks identified via a first-pass Bai-Perron residual audit (15% trimming), distinguished from impulse outlier controls $D_{T_b}^{\text{imp}} = \mathbb{I}(t = T_b)$.